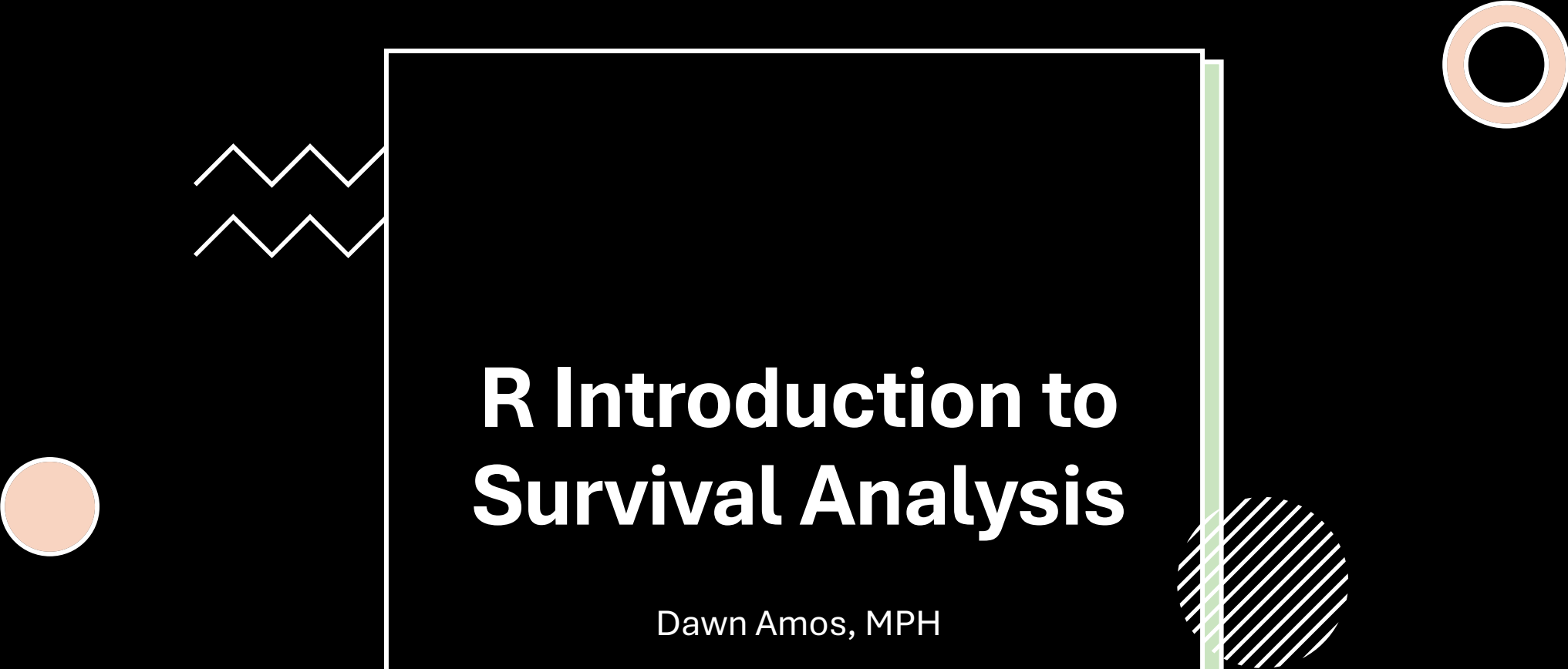


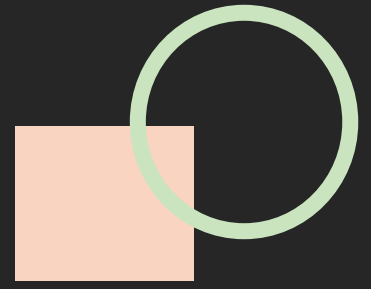


# **R Introduction to Survival Analysis**

Dawn Amos, MPH

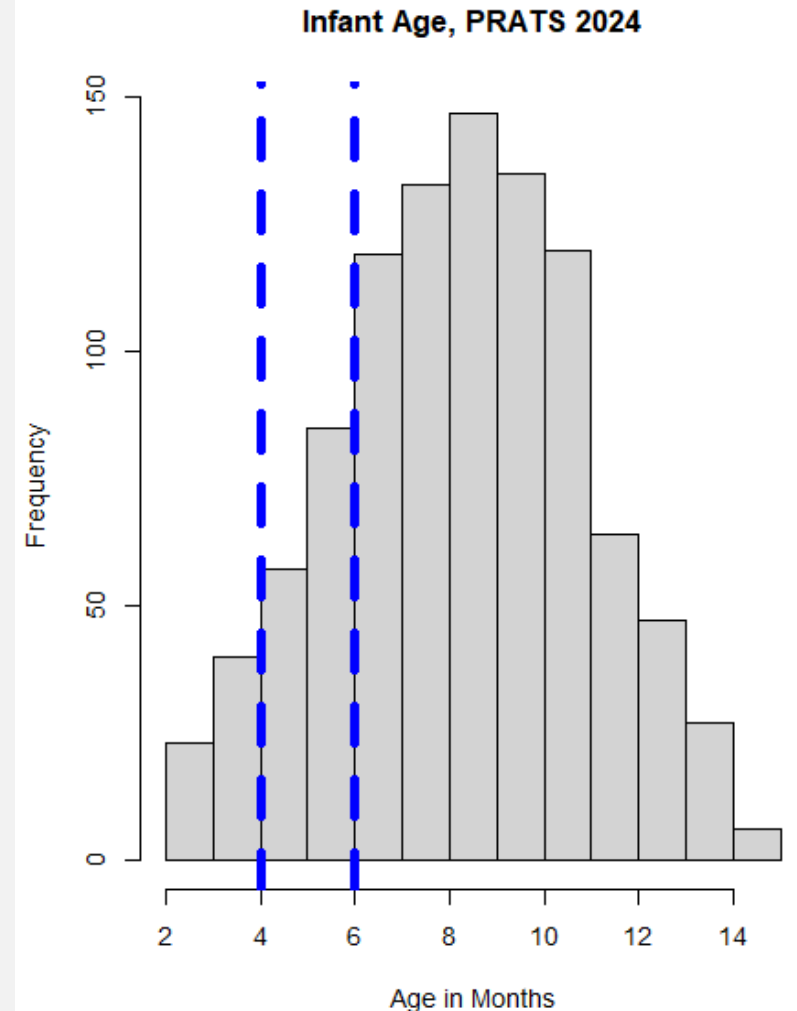
Idaho Department of Health and Welfare

# The Problem: Infant Feeding

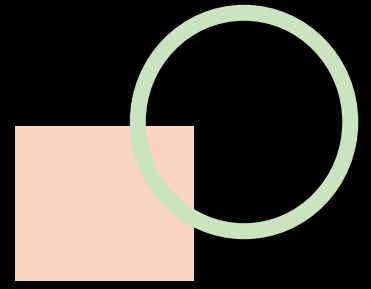


**Study Question:** What predicts when infants are introduced to solid foods?

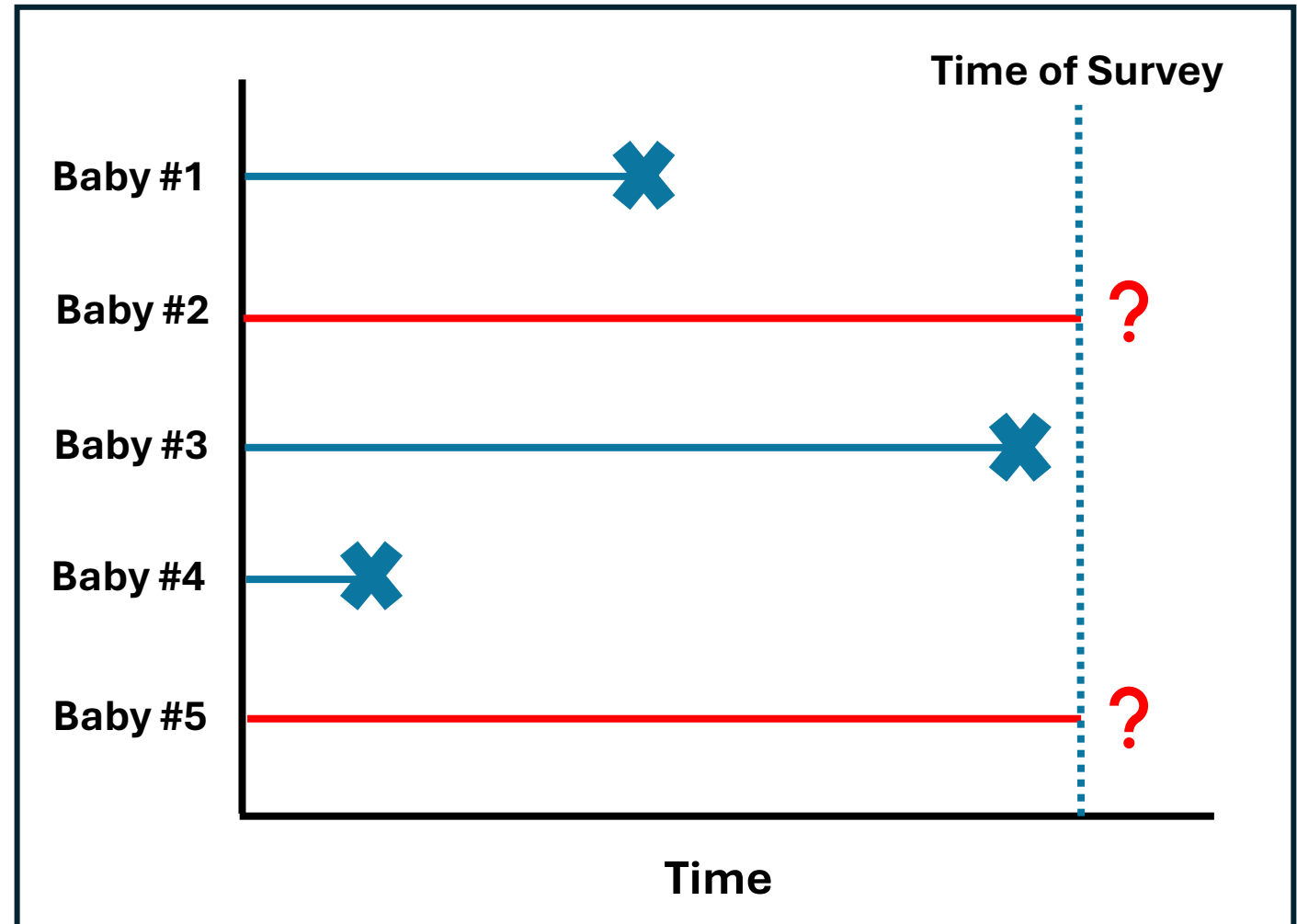
- Solids usually introduced at 4 – 6 months
- Sample
  - n = 877 (complete responses)
  - **Infant ages:** 2.6 to 15 months
  - **Never fed solids:** ~18% of sample (n = 154)



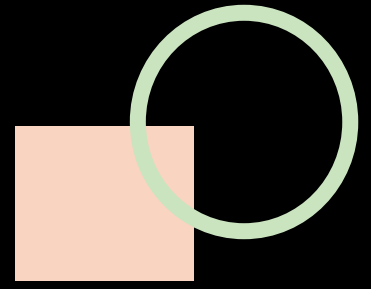
# The Problem: Infant Feeding, cont.



- **Issue:** 18% of our sample contains censored data
- **Censored data:** exact outcome is unknown
  - Outcome occurs outside of the study window or outcome is loss to follow-up



# The Problem: Infant Feeding, cont.



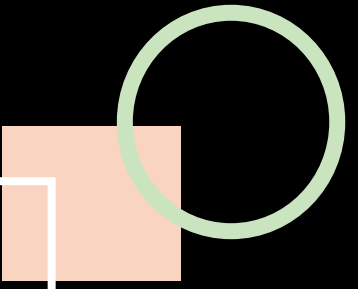
**What do we use when we...**

- Want to study what predicts when an event happens
- Have censored observations (unknown outcomes)

---

Answer:

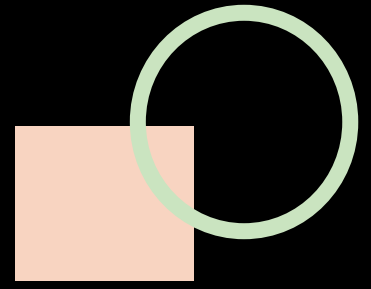
**Survival (Time-to-Event) Analysis**



# Survival Analysis

1. What is a survival analysis?
2. Test and model assumptions
3. How to prep your data
4. Different types of survival analysis

# Intro to Survival Analysis



**Survival Analysis:** analyzes the time to an event

Allows for the inclusion of censored data

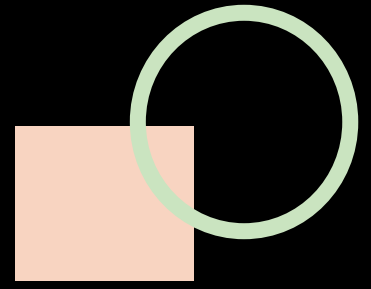
- Unknown outcomes don't need to be removed
- Censored data not required

Most common survival analyses:

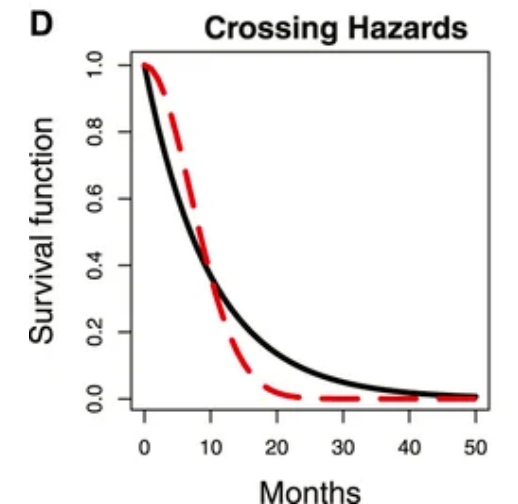
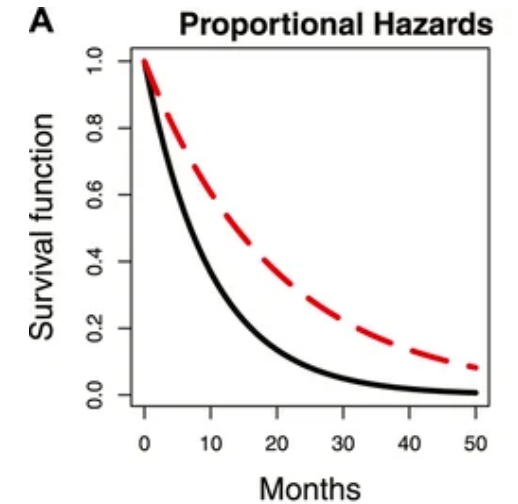
- **Kaplan-Meier method**
- **Log-Rank Test**
- **Cox proportional hazards model**



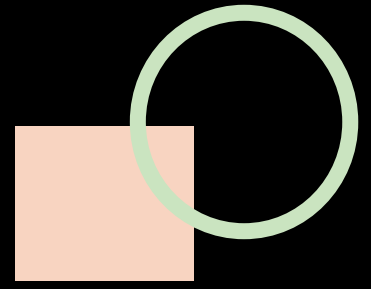
# Assumptions



- 1. Independent observations** → survival of one observation is independent from the survival of other observations
- 2. Independent censoring** → reason for a subject being censored is unrelated to the event
- 3. Proportional hazards** → hazard ratio between two groups are constant over time



# Prepping Your Data



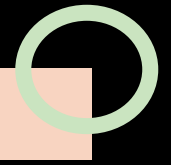
Using the R **survival** package

Two columns for each respondent:

- **Status** → binary variable (0-1) that indicates whether the event has happened to a respondent
- **Time**
  - If 0 ... time at study end
  - If 1 ... time of event



# Prepping Your Data, cont.



Example:

| Respondent | Fed solids | Infant age at study end (months) | Solid intro (months) | <u>Status</u> | <u>Time</u> |
|------------|------------|----------------------------------|----------------------|---------------|-------------|
| 101        | Yes        | 9                                | 6                    | 1             | 6           |
| 102        | No         | 2.5                              | -                    | 0             | 2.5         |
| 103        | Yes        | 7                                | 4.5                  | 1             | 4.5         |
| 104        | No         | 5                                | -                    | 0             | 5           |

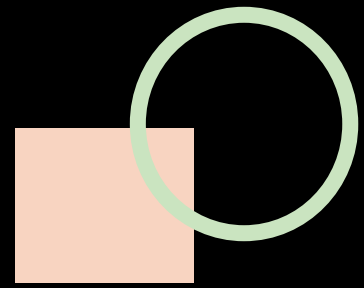
Green highlight: censored observation

Yellow highlight: complete observation

Practice Question:

| Respondent | Fed solids | Infant age at study end (months) | Solid intro (months) | <u>Status</u> | <u>Time</u> |
|------------|------------|----------------------------------|----------------------|---------------|-------------|
| 105        | No         | 4                                | -                    |               |             |
| 106        | No         | 3.5                              | -                    |               |             |
| 107        | Yes        | 8                                | 6.5                  |               |             |
| 108        | Yes        | 10                               | 5                    |               |             |

# Prepping Your Data, cont.



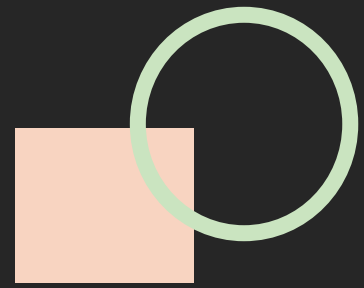
## Practice Answer:

| Respondent | Fed solids | Infant age at study end (months) | Solid intro (months) | Status | Time |
|------------|------------|----------------------------------|----------------------|--------|------|
| 105        | No         | 4                                | -                    | 0      | 4    |
| 106        | No         | 3.5                              | -                    | 0      | 3.5  |
| 107        | Yes        | 8                                | 6.5                  | 1      | 6.5  |
| 108        | Yes        | 10                               | 5                    | 1      | 5    |

Green highlight: censored observation

Yellow highlight: complete observation

# Prepping Your Data, cont.



## Input

```
library(dplyr)

# creating our variables for survival analysis
solids = solids %>%
  mutate(
    status = introduced_solids,
    time = case_when(
      status == 0 ~ infant_age_months,
      status == 1 ~ age_solids_introduced_months))

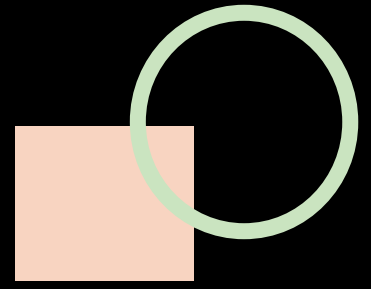
# checking our work
solids %>%
  select(infant_age_months,
         age_solids_introduced_months,
         time,
         status) %>%
  head()
```

## Output

```
Prep Variables, output
```

|   | infant_age_months | age_solids_introduced_months | time | status |
|---|-------------------|------------------------------|------|--------|
| 1 | 7.6               | 5.1                          | 5.1  | 1      |
| 2 | 6.9               | NA                           | 6.9  | 0      |
| 3 | 5.6               | NA                           | 5.6  | 0      |
| 4 | 4.5               | 4.5                          | 4.5  | 1      |
| 5 | 6.5               | 5.9                          | 5.9  | 1      |
| 6 | 6.1               | NA                           | 6.1  | 0      |

# Types: Kaplan-Meier

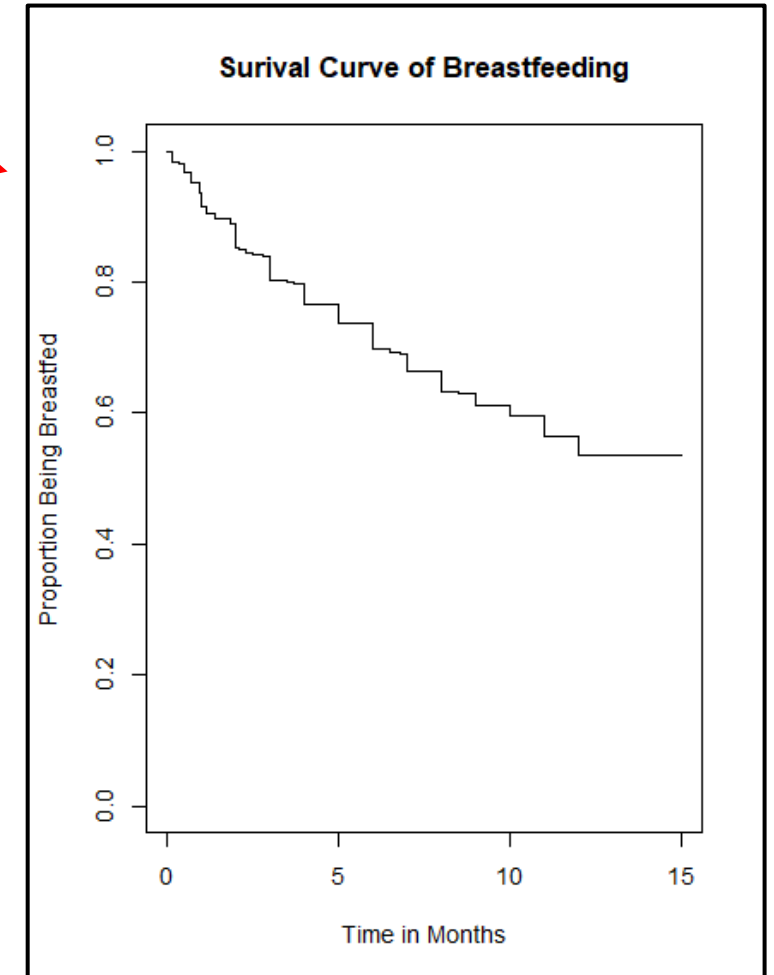


**Kaplan-Meier:** gives us a survival curve

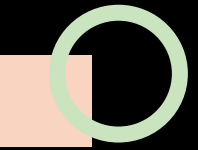
- Plots time on the x-axis and proportion of surviving observations on the y-axis

Tells us...

- Proportion of surviving observations at a certain time point
- Probability of survival at a certain time point



# Types: Kaplan-Meier, cont.



Kaplan Meier Curve

```
library(survival)

# Kaplan-Meier Curve
km_plot = survfit(Surv(time, status) ~ 1, data = solids)

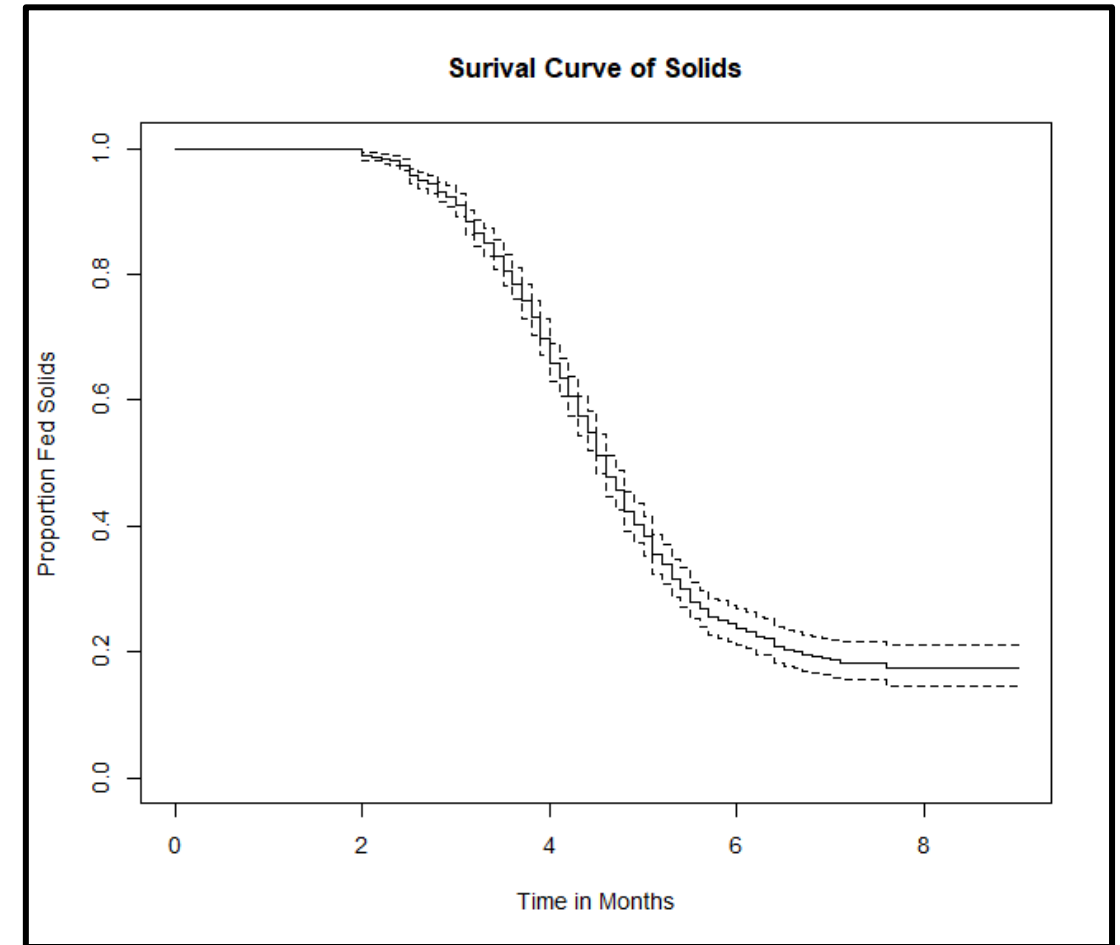
plot(km_plot,
     main = "Survival Curve of Solids",
     xlab = "Time in Months",
     ylab = "Proportion Fed Solids")

library(survey)
library(survminer)

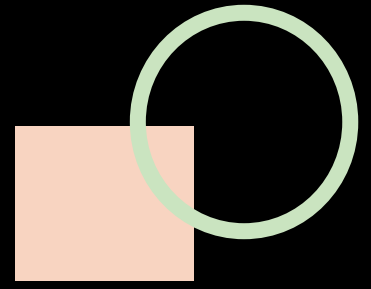
# creating a survey design
solids.design = svydesign(ids = ~1,
                        strata = ~ethnicity,
                        weights = ~survey_weight,
                        fpc = ~N,
                        data = solids)

# Kaplan-Meier Curve with a survey design (survey weights)
km_weighted_plot = svykm(Surv(time, status) ~ 1, design = solids.design)

plot(km_weighted_plot,
     main = "Survival Curve of Solids",
     xlab = "Time in Months",
     ylab = "Proportion Fed Solids")
```



# Types: Log-Rank Test

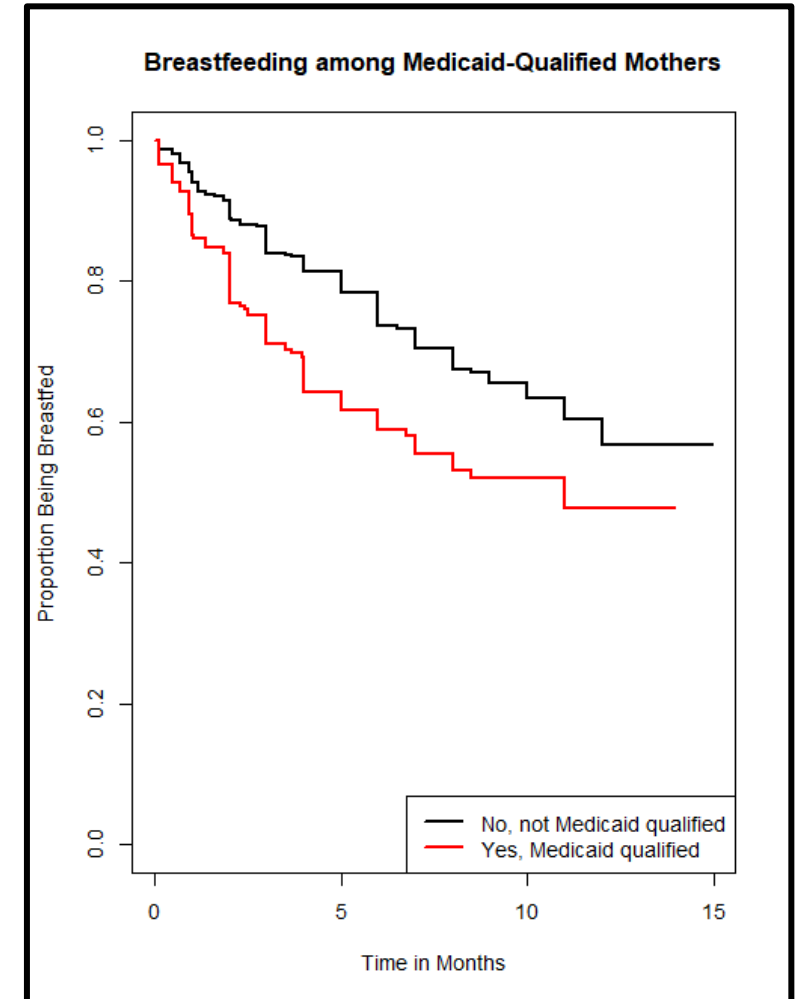


**Log-Rank Test:** compares the survival curve of two different groups

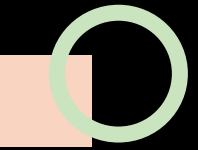
- Type of chi-squared test

Tells us...

- Are the survival curves of these two groups significantly different?



# Types: Log-Rank Test, cont.



## Input

```
Log-Rank

# Log-Rank Test
log_rank = survdiff(Surv(time, status) ~ insurance,
                    data = solids) %>%
  print()

# weighted Log-Rank Test
log_rank_weighted = svylogrank(Surv(time, status) ~ insurance,
                              design = solids.design) %>%
  print()
```

## Output

```
Log-Rank, output

> # Log-Rank Test
> log_rank = survdiff(Surv(time, status) ~ insurance,
+                     data = solids) %>%
+   print()
Call:
survdiff(formula = Surv(time, status) ~ insurance, data = solids)

              N Observed Expected (0-E)^2/E (0-E)^2/V
insurance=Private 472      337   356.1    1.0208    2.1066
insurance=Medicaid 414      288   291.8    0.0494    0.0867
insurance=None     114       95    72.1    7.2454    8.3896

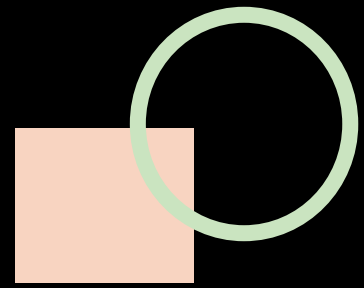
Chisq= 8.7 on 2 degrees of freedom, p= 0.01

>
> # weighted Log-Rank Test
> log_rank_weighted = svylogrank(Surv(time, status) ~ insurance,
+                               design = solids.design) %>%
+   print()
[[1]]
      score      se      z      p
[1,] -106.4852 261.4229 -0.4073292 0.683766188
[2,]  453.2192 155.8228  2.9085552 0.003631031

[[2]]
      Chisq      p
8.7599772 0.0125255

attr(,"class")
[1] "svylogrank"
```

# Types: Log-Rank Test, cont.

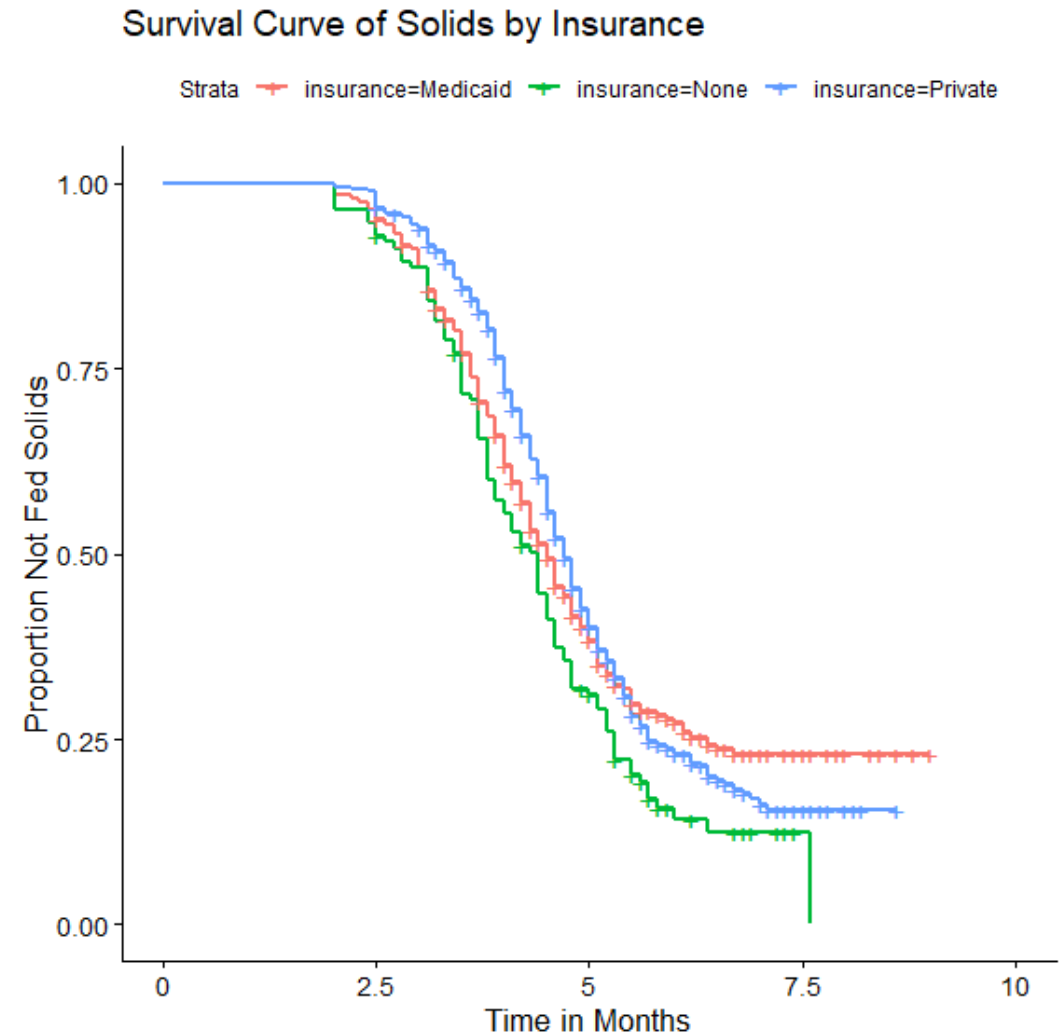


## Un-Weighted Data

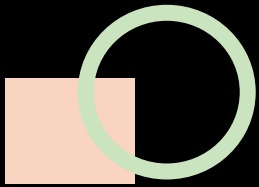
```
Plotting Log-Rank

# create fit first
insurance_fit = survfit(Surv(time, status) ~ insurance,
                        data = solids)

# using ggsurvplot()
ggsurvplot(insurance_fit,
            title = "Survival Curve of Solids by Insurance",
            xlab = "Time in Months",
            ylab = "Proportion Not Fed Solids")
```



# Types: Log-Rank Test, cont.

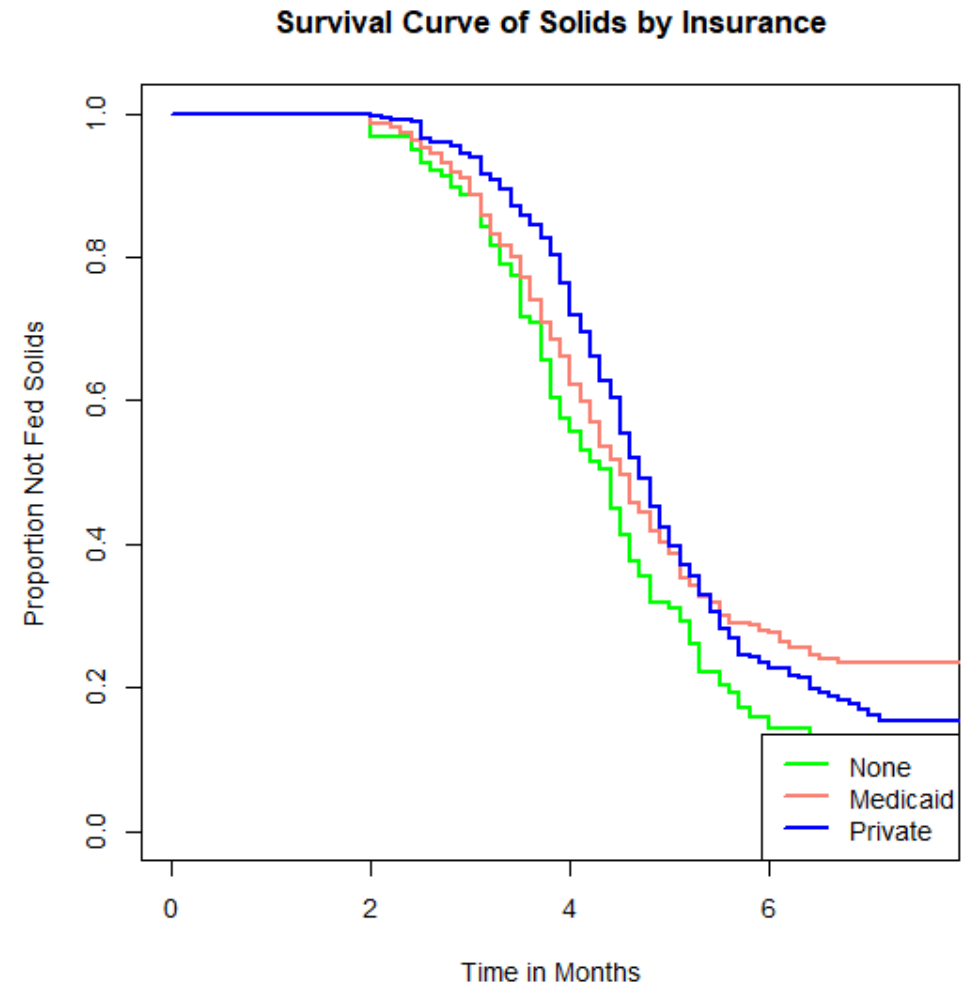


## Weighted Data

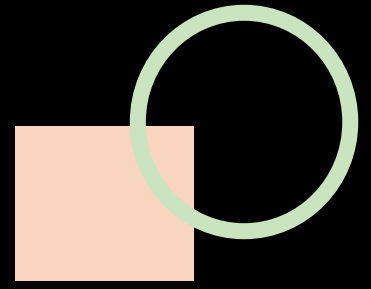
Plotting Log-Rank

```
# created weighted fit first
insurance_fit_weighted = svykm(Surv(time, status) ~ insurance,
                              design = solids.design)

# using plot()
plot(insurance_fit_weighted[["None"]],
     main = "Survival Curve of Solids by Insurance",
     xlab = "Time in Months",
     ylab = "Proportion Not Fed Solids",
     col = "green", lwd = 2)
lines(insurance_fit_weighted[["Medicaid"]], col = "salmon", lwd = 2)
lines(insurance_fit_weighted[["Private"]], col = "blue", lwd = 2)
# Add a legend
legend("bottomright",
     legend = c("None", "Medicaid", "Private"),
     col = c("green", "salmon", "blue"),
     lty = 1,
     lwd = 2)
```



# Types: Cox Proportional Hazards Model

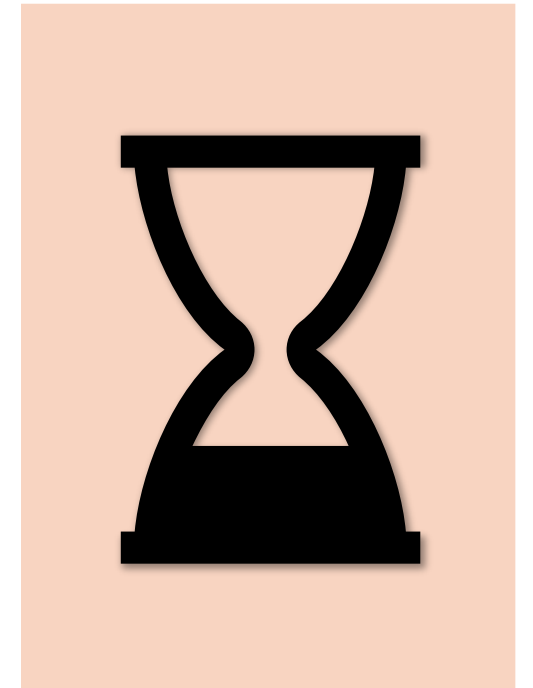


**Cox Proportional Hazards Model:** assesses how different predictors impact the time it takes for an event to occur

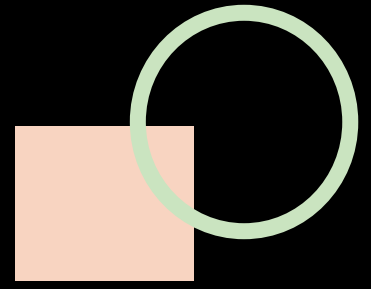
- Gives us hazard ratios

Tells us...

- Does this predictor speed-up or slow-down the rate of event occurrence?



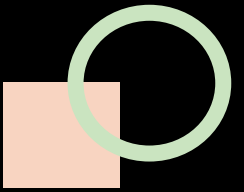
# Types: Cox Proportional Hazards Model



A ***hazard ratio*** is...

- A rate of occurrence
- Comparative → relative to the reference group
- Instantaneous → at given moment in the study period

# Types: Cox Proportional Hazards Model



Imagine we have two buildings.  
Each building contains with 1,000 lightbulbs.

## Building A

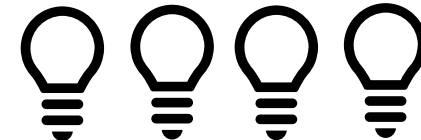
Each hour, 2 lightbulbs burn out.



Rate: 2 per hour

## Building B

Each hour, 4 lightbulbs burn out.



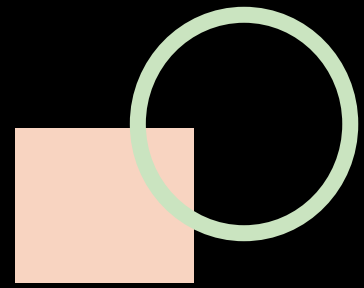
Rate: 4 per hour

## Hazard Ratio

Building B's **rate of occurrence** is 2 **times that of**  
**Building A's** rate of occurrence in **any given moment**.

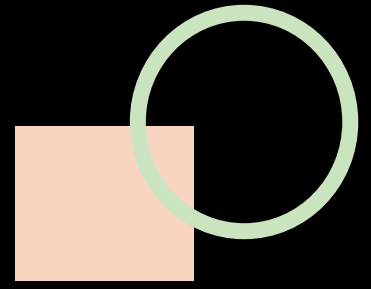
- **Rate of occurrence**
- **Relative to reference group**
- **At any given moment**

# Types: Cox Proportional Hazards Model



| Hazard Ratio (HR) Value | Interpretation                          | Example                                                                                         |
|-------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------|
| HR = 1.0                | No difference                           | “This group’s rate of occurrence at any given moment is the same as the reference group.”       |
| HR < 1.0                | Lower hazard → slower event occurrence  | “This group’s rate of occurrence is 0.5 times that of the reference group at any given moment.” |
| HR > 1.0                | Higher hazard → faster event occurrence | “This group’s rate of occurrence is 1.5 times that of the reference group at any given moment.” |

# Types: Cox Proportional Hazards Model



## Un-Weighted Data

### Input

```
Un-Weighted Cox Proportional Hazards Model

cox_model = coxph(Surv(time, status) ~ insurance +
ethnicity + maternal_age,
                  data = solids) %>%

print()
```

### Input

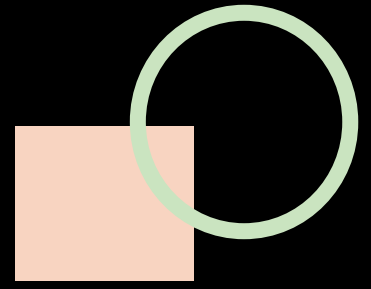
```
Un-Weighted Cox Proportional Hazards Model, output

> cox_model = coxph(Surv(time, status) ~ insurance + ethnicity +
maternal_age,
+                   data = solids) %>%
+   print()
Call:
coxph(formula = Surv(time, status) ~ insurance + ethnicity +
      maternal_age, data = solids)

              coef exp(coef) se(coef)      z      p
insuranceMedicaid  0.042084  1.042982  0.080584  0.522 0.60151
insuranceNone       0.345588  1.412820  0.116291  2.972 0.00296
ethnicityHispanic   0.058476  1.060219  0.089651  0.652 0.51423
maternal_age        -0.023096  0.977169  0.007532 -3.066 0.00217

Likelihood ratio test=17.64 on 4 df, p=0.001453
n= 1000, number of events= 720
```

# Types: Cox Proportional Hazards Model



## Weighted Data

### Input

```
Weighted Cox Proportional Hazards Model

cox_model_weighted = svycoxph(Surv(time, status) ~ insurance +
  ethnicity + maternal_age,
                             design = solids.design) %>%
  print()
```

### Input

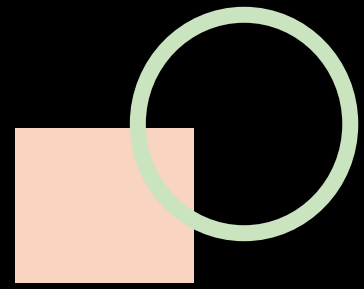
```
Weighted Cox Proportional Hazards Model, output

> cox_model_weighted = svycoxph(Surv(time, status) ~ insurance +
  ethnicity + maternal_age,
  +                             design = solids.design) %>%
  +   print()
Call:
svycoxph(formula = Surv(time, status) ~ insurance + ethnicity +
  maternal_age, design = solids.design)

              coef exp(coef)  se(coef) robust se      z      p
insuranceMedicaid  0.031373  1.031870  0.080608  0.080293  0.391 0.69600
insuranceNone       0.338935  1.403452  0.116407  0.114642  2.956 0.00311
ethnicityHispanic   0.057604  1.059295  0.094099  0.088061  0.654 0.51302
maternal_age        -0.023337  0.976933  0.007531  0.007848 -2.974 0.00294

Likelihood ratio test=  on 4 df, p=
n= 1000, number of events= 720
```

# Types: Cox Proportional Hazards Model



```
Weighted Cox Proportional Hazards Model, output

> cox_model_weighted = svycoxph(Surv(time, status) ~ insurance +
+   ethnicity + maternal_age,
+   design = solids.design) %>%
+   print()
Call:
svycoxph(formula = Surv(time, status) ~ insurance + ethnicity +
  maternal_age, design = solids.design)
```

|                   | coef      | exp(coef) | se(coef) | robust se | z      | p       |
|-------------------|-----------|-----------|----------|-----------|--------|---------|
| insuranceMedicaid | 0.031373  | 1.031870  | 0.080608 | 0.080293  | 0.391  | 0.69600 |
| insuranceNone     | 0.338935  | 1.403452  | 0.116407 | 0.114642  | 2.956  | 0.00311 |
| ethnicityHispanic | 0.057604  | 1.059295  | 0.094099 | 0.088061  | 0.654  | 0.51302 |
| maternal_age      | -0.023337 | 0.976933  | 0.007531 | 0.007848  | -2.974 | 0.00294 |

```
Likelihood ratio test= on 4 df, p=
n= 1000, number of events= 720
```

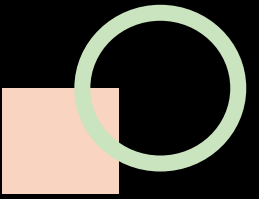
Number of non-censored observations

**Model:** predicting time of solid food introduction by insurance type, ethnicity, and maternal age

## Findings:

- At any given moment, mothers with no insurance introduced solids at a rate that is 1.4 times that of mothers with private insurance ( $p < 0.01$ ).
  - Faster time-to-event
- For every year increase in age, the rate of introducing solids is 0.98 times that of mothers one year younger ( $p < 0.01$ ).
  - Slower time-to-event

# Recap of Content



- **Survival analysis** analyzes the time to an event.
- Survival analyses can handle censored (incomplete) observations.
- To use R's `survival` package, you need to create a status and time variable.
- The three most common forms of survival analysis are...
  - **Kaplan-Meier Method:** gives you a survival curve
  - **Log-Rank Test:** compares the survival curve of different groups to determine statistical significance
  - **Cox Proportional Hazards Model:** allows you to see how certain predictors speed up or slow down the rate of occurrence for an event
    - **Hazard ratios** are relative rates of occurrence for a any single unit of time

**Thanks for listening!**